

Factor of Safety

Electric Jet-Driven Surfboard

*Fatigue analysis of the composite shell under combined bending and torsion.
Conservative assumptions throughout to cover the gap in published fiberglass/epoxy fatigue data.*

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THE OPPORTUNITY

A high-performance electric watercraft — and a data gap to engineer around.

Jet-driven electric surfboards deliver real propulsion loads on a composite hull. Published fatigue data on fiberglass/epoxy is scarce.

MARKET CONTEXT

Emerging electric watercraft

Battery-electric jetboards are a growing recreational segment. Structural reliability is table stakes — a hull fatigue failure is a safety event.

ENGINEERING CHALLENGE

Combined loads on a composite shell

The hull sees fully-reversed bending from rider + board weight plus steady torque from the driveshaft — classic fatigue territory.

THE DATA GAP

Fiberglass/epoxy fatigue data is thin

Unlike steel, composite S-N curves and notch-sensitivity data for this exact layup are not readily published. That shapes every assumption we made.

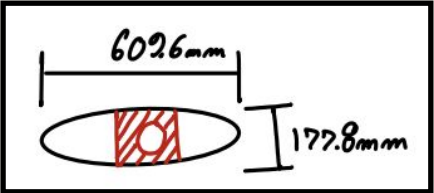
TALK TRACK | "Our job wasn't just to run the math. It was to run the math responsibly in a space where the textbook constants were never built for this material."

DESIGN OVERVIEW

The board: 2.4 m composite shell, rider + propulsion unit.

Geometry and loads that drive every downstream calculation.

BOARD LENGTH	2,413 mm	<i>≈ 95 in</i>
CROSS-SECTION	609.6 × 177.8 mm	<i>modeled as solid rectangle</i>
RIDER + BOARD MASS	86 kg	<i>844 N — split 422 N / 422 N</i>
PROPULSION UNIT MASS	34 kg	<i>334 N, fixed mount</i>
MOTOR PEAK POWER	25 kW	<i>~33 hp, 9,000 rpm</i>
OPERATING POINT	90% of peak	<i>22.5 kW at 8,100 rpm</i>



LOADING & STRUCTURAL MODEL

Simply-supported beam, two point loads, superposition.

Shigley Table A-9 cases combined to find the critical bending moment.

MAX BENDING MOMENT

477 N·m

Occurring at the rider load location ($a_1 = 804$ mm).

MAX DEFLECTION

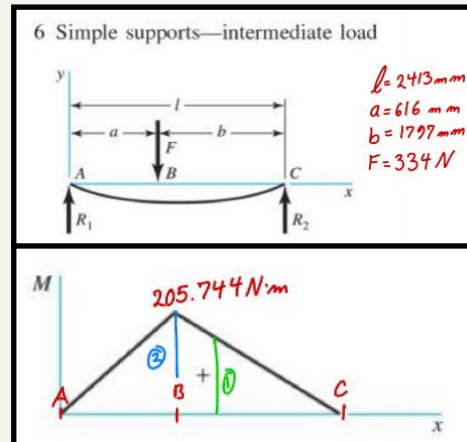
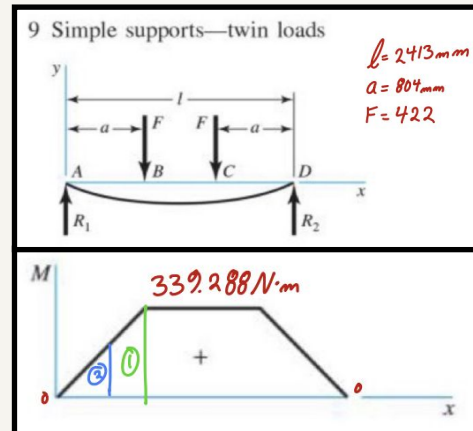
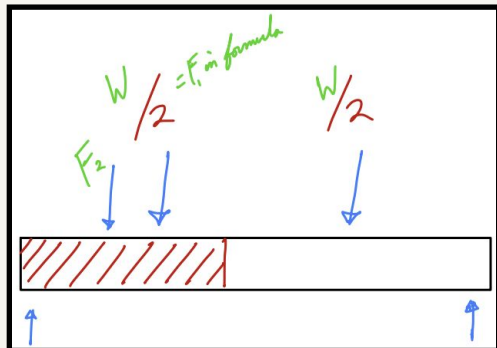
0.0737 mm

Negligible relative to board length — stiffness is not the limit.

DRIVESHAFT TORQUE

26.53 N·m

Steady torque at 22.5 kW, 8,100 rpm (90% of peak).



KEY RELATIONS

$$I = (1/12) b h^3 = 2.86 \times 10^{-4} \text{ m}^4$$

$$M_1 @ 804 \text{ mm} = 476.5 \text{ N·m} \leftarrow \text{governing}$$

$$M_2 @ 616 \text{ mm} = 413.2 \text{ N·m}$$

$$T = P \cdot 60 / (2\pi N) = 26.53 \text{ N·m}$$

MATERIAL TESTING

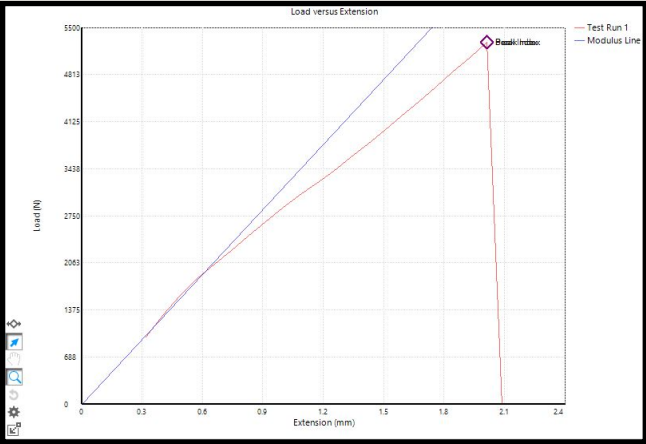
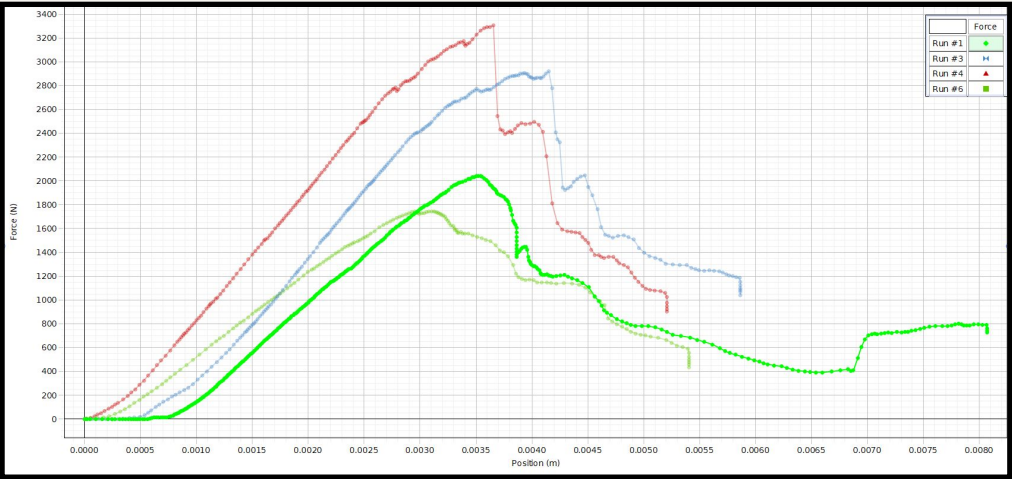
We tested the actual layup — not a textbook constant.

Four-point bending and tensile tests established the baseline strength we fed into the fatigue model.

ULTIMATE TENSILE STRENGTH

$S_{ut} =$
91.93 MPa

Measured value — the single most important input to the entire Marin endurance calculation.



THE CONSERVATIVE APPROACH

Three deliberate de-rates to cover the composite data gap.

Each assumption reduces usable strength on purpose — the margin we calculate is a floor, not a ceiling.

ASSUMPTION	STANDARD / TEXTBOOK	OUR (CONSERVATIVE) CHOICE	WHY IT HELPS
Endurance limit S'_e	$0.5 \cdot S_{ut}$ (steel convention)	$0.3 \cdot S_{ut} = 27.58 \text{ MPa}$	Composite S-N curves aren't published — we assume a lower ceiling than steel.
Notch sensitivity q	$q < 1$ when data allows relief	$q = 1$ (fully notch-sensitive)	No composite notch data — so we never discount the stress concentration.
Surface factor k_a	k_a computed from $S_{ut} \rightarrow 1.13$	k_a capped at 0.9 (machined)	Physically $k_a \leq 1.0$; we further penalize to reflect real surface variability.
Size factor k_b	d range of $51 \leq d < 254 \text{ mm}$ $d_e \rightarrow 266 \text{ mm}$	d capped at 254 mm (upper limit)	No composite notch data — so we never discount the stress concentration.

Stacking these choices means the calculated factor of safety is a lower bound. Real margin is very likely larger.

MARIN FACTORS · ENDURANCE LIMIT

Modifying S'_e to reflect the real component — not a polished lab coupon.

Every k-factor earns its keep; none are optimistic.

k_a
0.90

Surface

Machined finish (capped at ≤ 1.0)

k_b
0.633

Size

Evaluated at d at 254 mm upper bound

k_c
1.00

Loading

Bending

k_d
1.00

Temperature

No data for composite at op. temps

k_e
0.897

Reliability

90% design reliability target

$$S_{e'} = 0.3 \cdot S_{ut} = 27.58 \text{ MPa}$$

$$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot S'_{e'}$$

$$= (0.9)(0.633)(1.0)(1.0)(0.897)(27.58 \text{ MPa})$$

MODIFIED ENDURANCE LIMIT

$$S_e = 14.09 \text{ MPa}$$

RESULTS · FACTOR OF SAFETY

Two independent criteria agree: there is massive margin.

Both the DE-Gerber and DE-Goodman approaches return $n \approx 22$ — more than 20× the minimum.

DE-GERBER

$$n = 22.21$$

Distortion Energy — Gerber criterion

Mean + alternating stress handled separately; Gerber is the standard first-pass for a new ductile design.

DE-GOODMAN

$$n = 21.82$$

Distortion Energy — Modified Goodman

More conservative linear combination. Nearly identical result confirms the answer is robust.

MINIMUM ACCEPTABLE

$$n = 1.0$$

TYPICAL PROTOTYPE TARGET

$$n \approx 2-3$$

OUR RESULT

$$n \approx 22$$

*$\approx 10\times$ the prototype bar,
with the data gap priced in.*

THE TAKEAWAY

Why this design is safe — despite the composite data gap.

01

The margin is big — twice over.

Two independent fatigue criteria (Gerber and Modified Goodman) agree within 2%. $n \approx 22$ against a design minimum of 1.0. The result is not criterion-dependent.

02

The margin is conservative — on purpose.

S_e at $0.3 \cdot S_{ut}$ (not 0.5), full notch sensitivity, machined surface penalty, size factor at its bound. Every assumption removes strength the material likely has.

03

Real-world margin is almost certainly larger.

When experimental S-N data for this specific fiberglass/epoxy layup becomes available, the calculated n will go up — not down. This is the floor.

NEXT STEPS

From conservatively safe to optimally designed.

The same margin can be redirected from safety reserve into lighter, stiffer, cheaper hulls — once we have the data.

01

Fatigue S-N testing of this layup

Replace the 0.3·Sut assumption with measured endurance data — the single largest conservatism in the analysis. (Stress-Number of cycles Testing)

02

Layup / ply / fiber-orientation matrix

Characterize mechanical sensitivity across configurations to enable structured design optimization.

03

Impact and shear testing

Address failure modes that four-point bending and tension do not capture — directly relevant to hydrodynamic impact.

04

Notch sensitivity characterization

Drop $q = 1$ once composite-specific data exists; expected to further increase calculated margin.

THANK YOU

Questions?

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Machine Design · EML 3500

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